Supplementary material for "The great tit HapMap project: a continental-scale analysis of genomic variation in a songbird"

Table S1 Sampling details for the 28 European great tit populations.

Population	Region	N	Latitude	Longitude
Antwerp_Belgium	Belgium	15	51.226	4.390
Bulgaria	Balkans	3	42.852	24.986
Cambridge_UK	England	28	52.198	0.152
Crete	Crete	4	35.183	24.960
Font_Roja_Spain	Spain	19	38.664	-0.544
Gotland_Sweden	Sweden	32	57.504	18.466
Harjavalta_Finland	Finland	44	61.314	22.139
Italy	Italy	13	42.386	13.354
La_Rouviere_France	France	19	43.930	4.235
Loch_Lomond_Scotland	Scotland	21	56.108	-4.562
Mariola_Spain	Spain	21	37.717	-0.658
Montpellier_France	France	36	43.098	3.569
Oulu_Finland	Finland	29	65.013	25.468
Pilis_Mountains_Hungary	Hungary	28	47.733	18.917
Pirio_Muro_Corsica	Corsica	27	42.110	9.104
Radolfzell_Germany	Germany	18	47.759	8.957
Romania	Balkans	7	46.497	24.582
Sardinia	Sardinia	9	40.056	9.025
Seewisen_Germany	Germany	30	47.978	11.233
Tartu_Estonia	Estonia	32	58.375	26.739
Turkey	Turkey	11	39.464	34.797
Velky_Kosir_Czech_Republic	Czech Republic	22	49.547	17.062
Vienna_Austria	Austria	27	48.239	16.347
Vlieland_NL	Netherlands (Vlieland)	15	53.238	4.932
Westerheide	Netherlands	50	52.250	5.190
Wytham_UK	England	48	51.776	-1.303
Zurich_Switzerland	Switzerland	22	47.383	8.523
Zvenigorod_Russia	Russia	17	55.738	36.857

Table S2 Details of outlier F_{ST} regions in European great tit populations. For each region, F_{ST} was calculated for each population versus Turkey. The "number of outliers represents" the number of *populations* in which each region was detected as an outlier. All genes found within 500kb outlier windows are displayed.

Chromosome	Position (MB)	Number of outliers	Outlier comparisons	Genes
2	81 5	12	Austria, Belgium, Czech Republic, Estonia, Finland, France Germany, Hungary, Netherlands, Scotland, Sweden, Switzerland	PADP7, SRDSA1, NSUN2, UBE2QL, MED10
6	18	10	Belgium, Corsica, England, France, Germany, Hungary, Italy Netherlands, Russia, Sweden	RASGEF1A, RET, CSGALNACT2
3	28 5	2	England, Spain	PPP1CB, CALM2, TTC7A, MCFD2
1A	70 5	2	Czech Republic, Russia	CDKN1B
4A	12	2	England, Scotland	COL4A5
6	8	2	Finland, Scotland	SLC18A3, RGR, LRI1, LRI2, CDHR1, GHITM
3	28	1	Spain	MRPL33, RBKS, BRE, FOSL2
1A	48 5	1	Finland	CBX6, CBX7, DNAL4, SUN2, NPTXR, JOSD1, TOMM22, KDELR3, DMC1, GTPBP1, DDX17, CBY1 KCNJ4, THEM184B, CSNK1E, PLA2G6, MAFF, BAIAP2L2, SLC16A8, PICK1, SOX10, MICALL1 EIF3L, ANKRDS4, GCAT, POLR2F
6	7 5	1	Scotland	BMPR1A, SNGC, MMRN2, GLUD1, FAM35A, GPRIN2, PARG, NCOA4, OGDHL, CHAT
8	7 5	1	Vlieland	-
8	8	1	Belgium	-

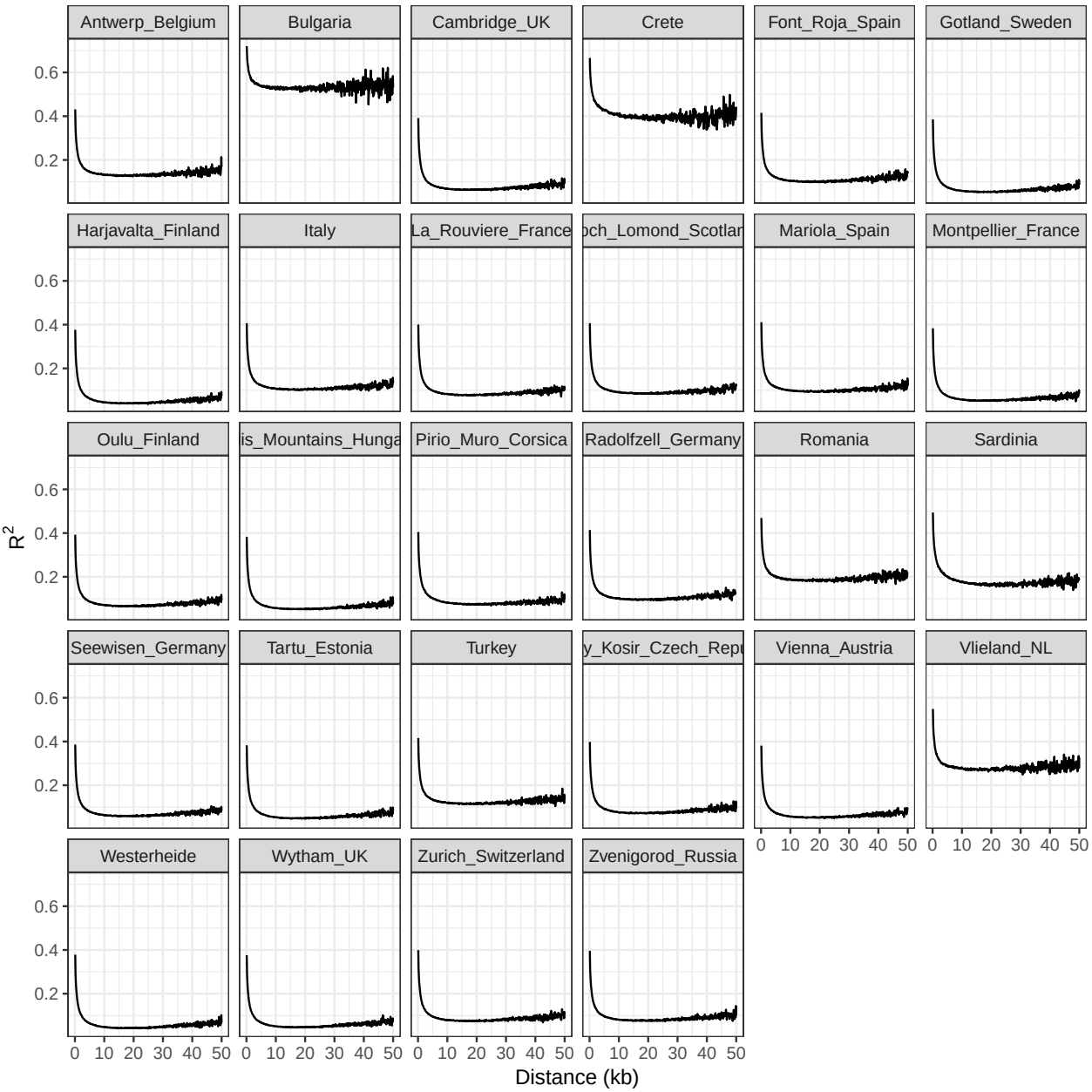

Figure S1 Linkage disequilibrium in European great tit populations. Lines are means of R^2 values for all pairs of markers within 100bp distance bins.

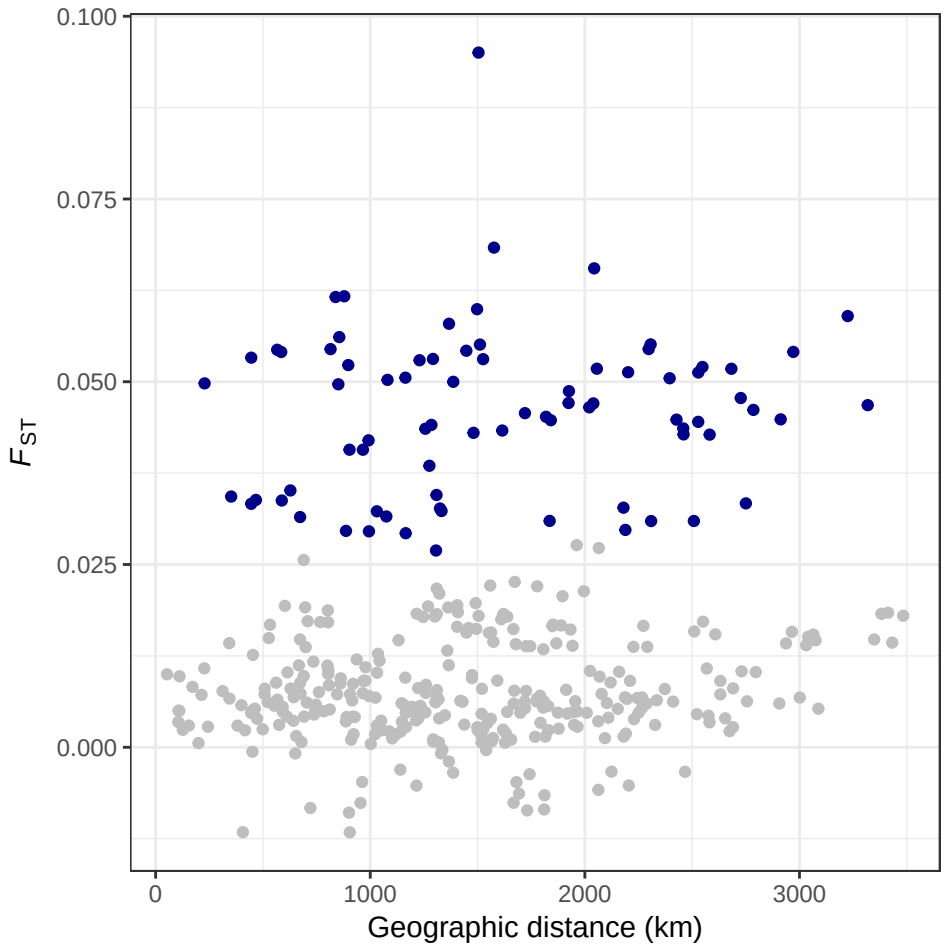

Figure S2 Pairwise F_{ST} in relation to geographic distance among European great tit populations. Pairwise comparisons involving the island populations of Corsica, Sardinia and Crete are coloured in dark blue.

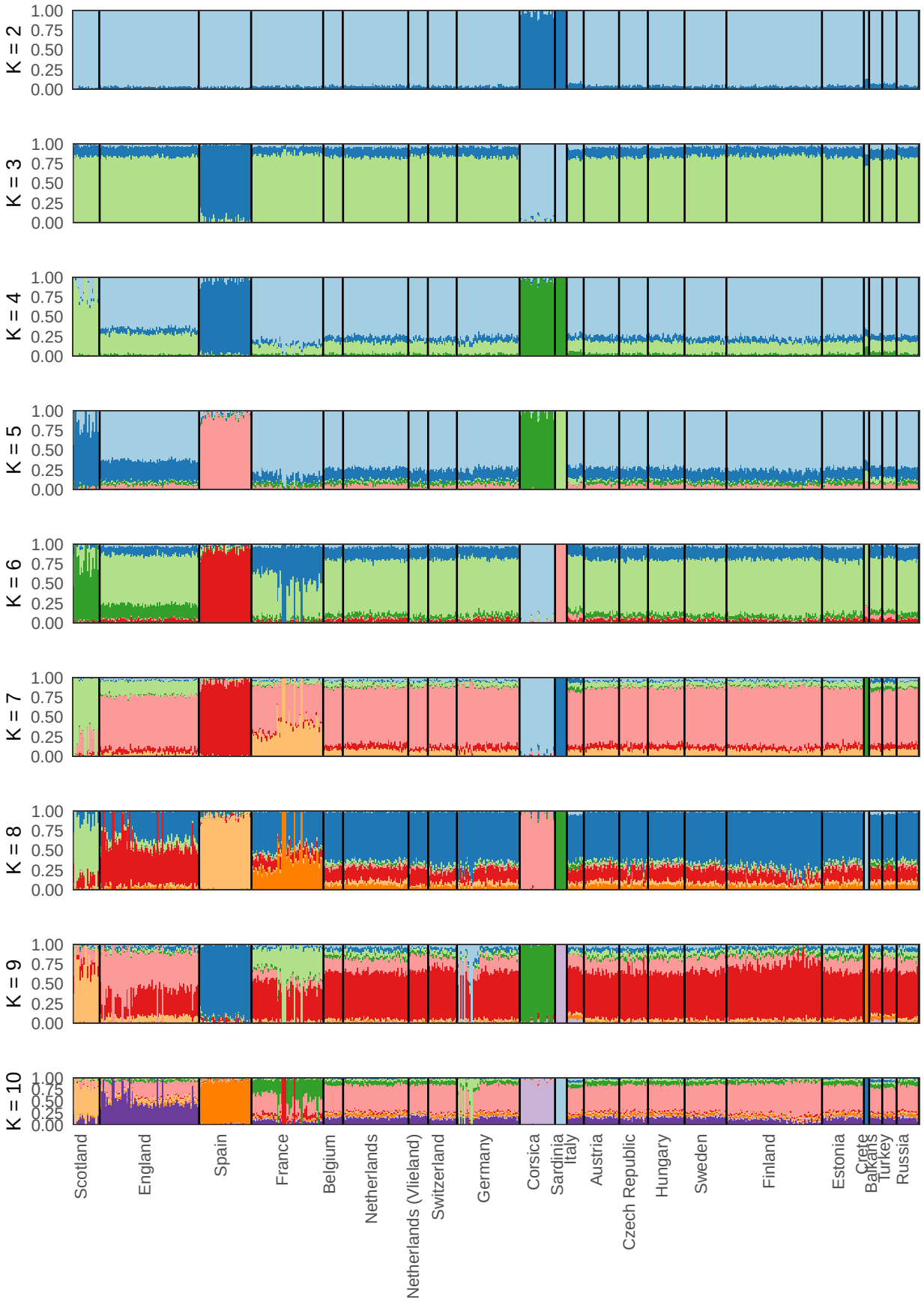
K = 2
K = 3
K = 4
K = 5
K = 6
K = 7
K = 8
K = 9
K = 10
1.00
0.75
0.50
0.25
0.00
Scotland
England
Spain
France
Belgium
Netherlands
Netherlands (Vlieland)
Switzerland
Germany
Corsica
Sardinia
Italy
Austria
Czech Republic
Hungary
Sweden
Finland
Estonia
Crete
Balkans
Turkey
Russia

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Figure S3 Admixture analyses at $K = 2$ to 10 in European Great tit populations.

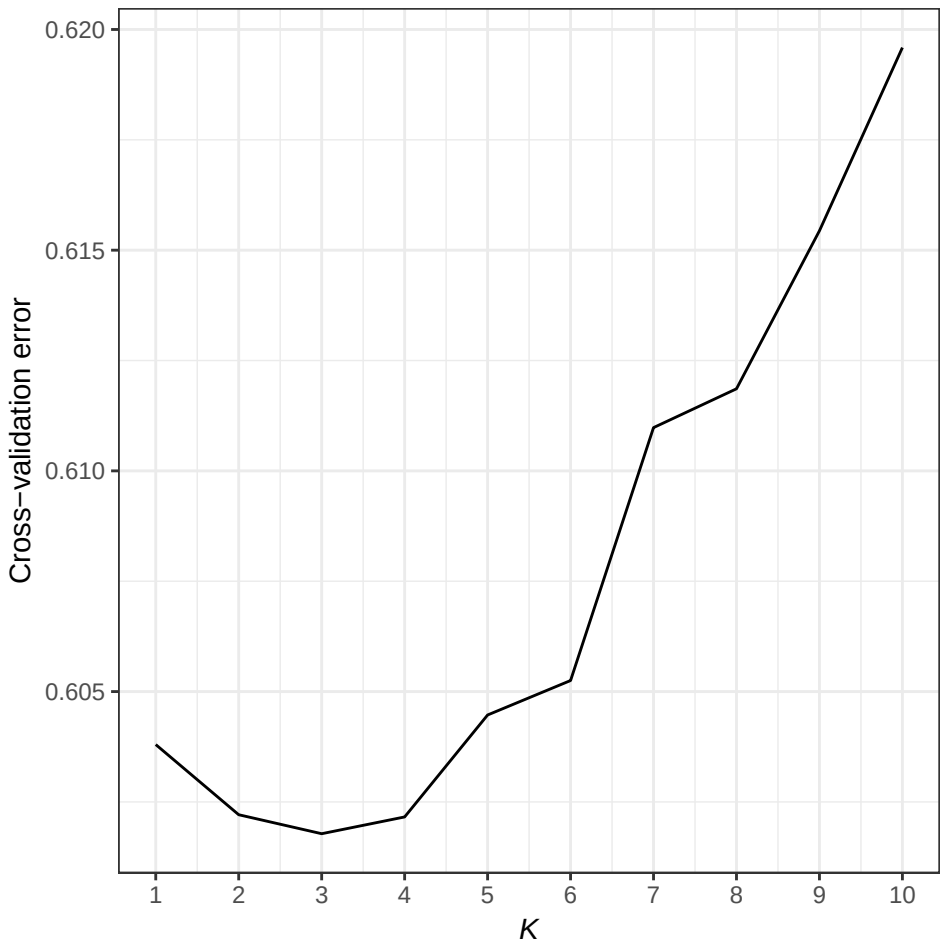

Figure S4 Cross-validation error from admixture analyses at $K = 1$ to 10.

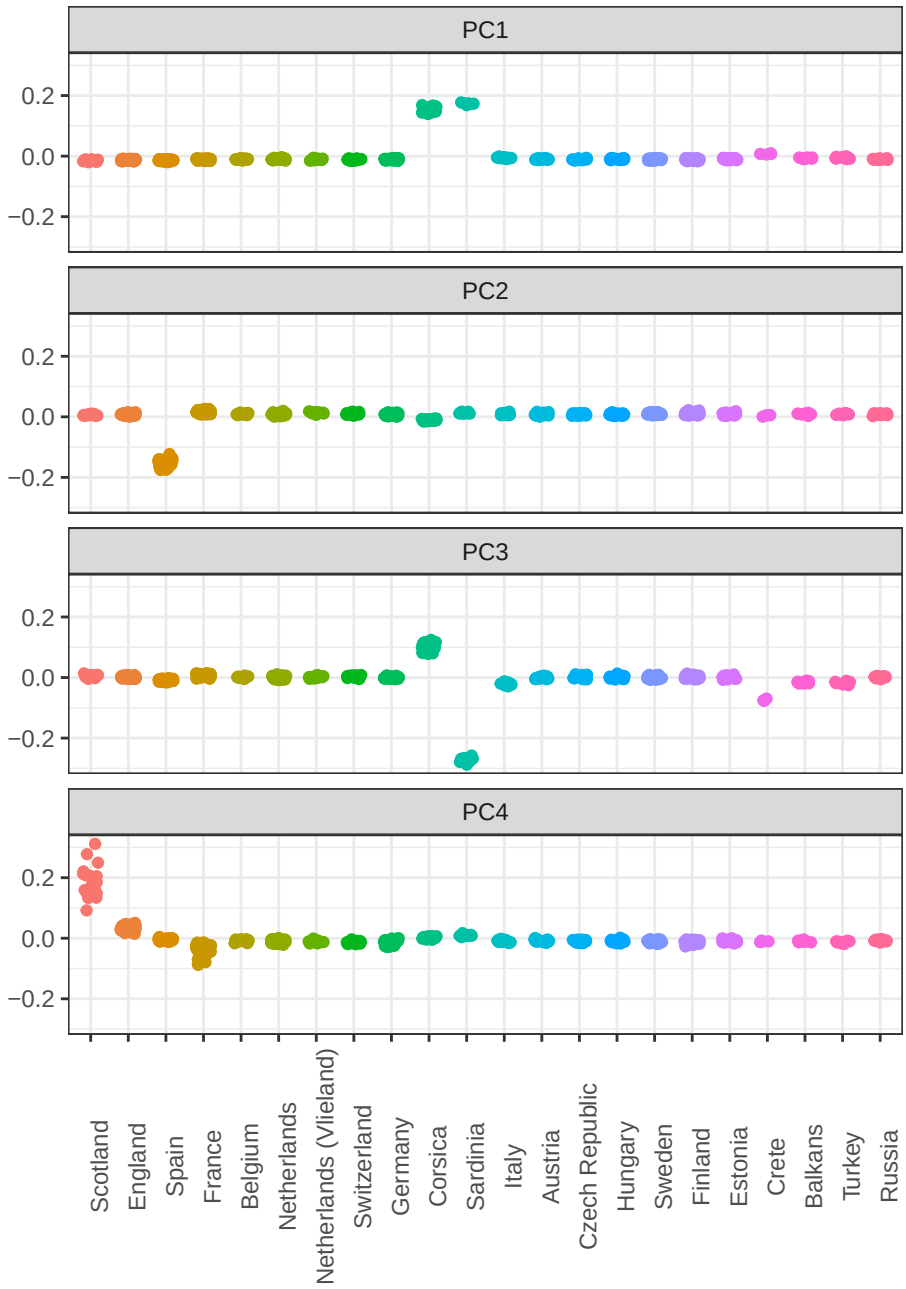

Figure S5 Results from PCA analysis of European great tit populations.

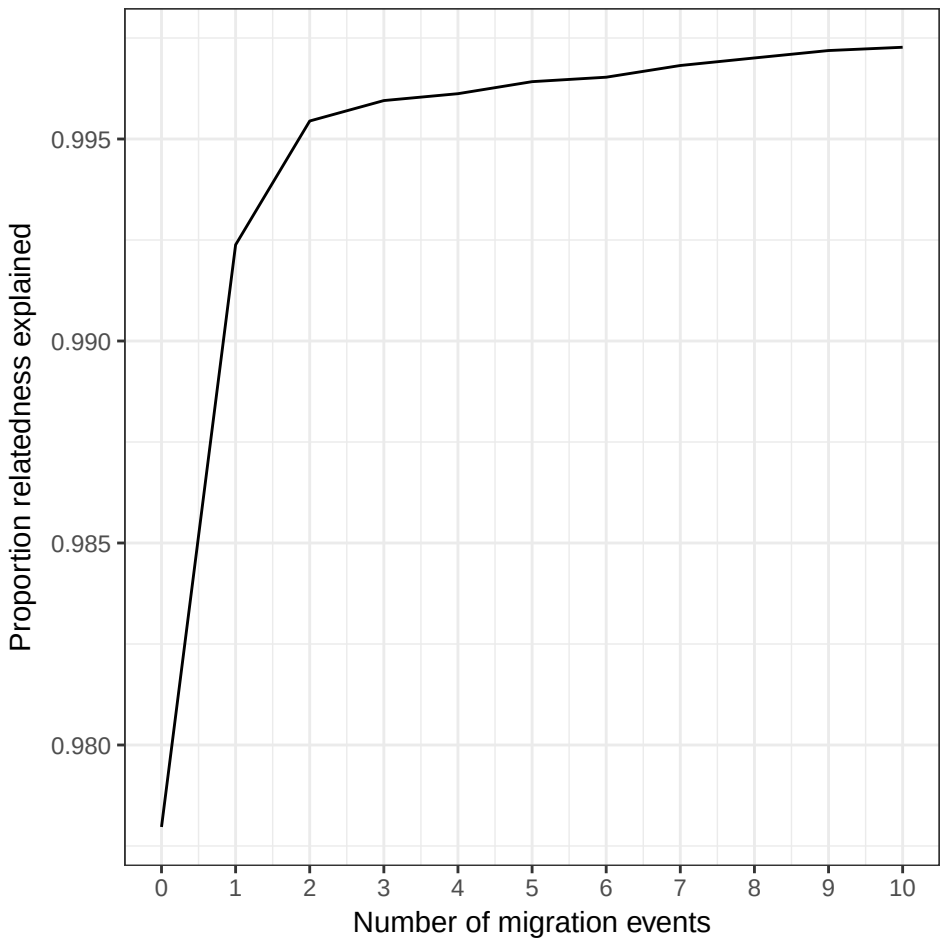

Figure S6 Proportion of relatedness explained among populations in relation to the number of migration events allowed in Treemix analyses.

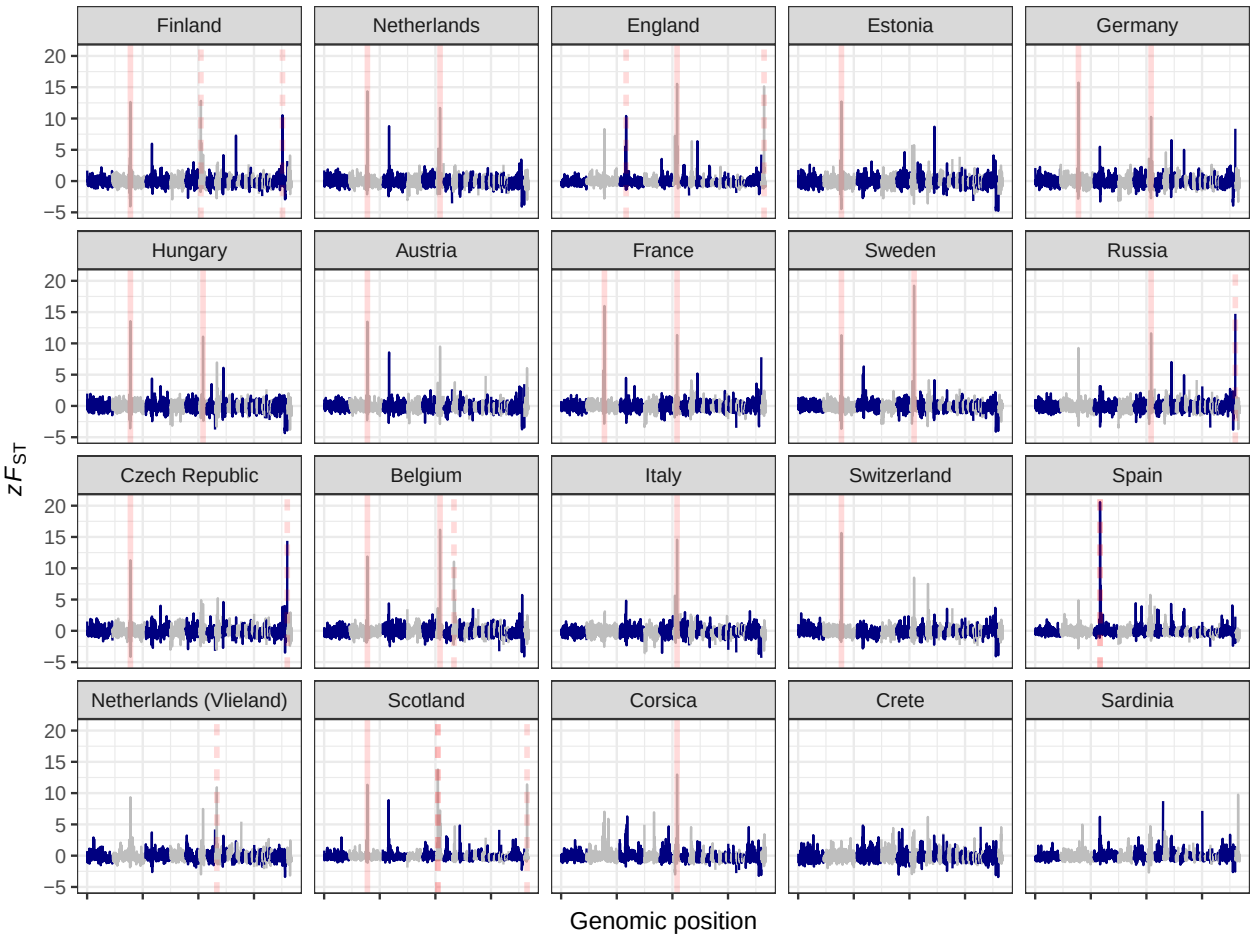

Figure S7 Landscapes of relative genomic differentiation in European great tit populations. zF_{ST} across the genome is averaged in 500kb windows, with each panel displaying a pairwise comparison between with the proposed refugial population in Turkey. Red lines represent F_{ST} outliers (windows with mean $zF_{ST} > 10$) shared across more than two comparisons (solid red lines), or unique to one or two comparisons (dashed red lines).

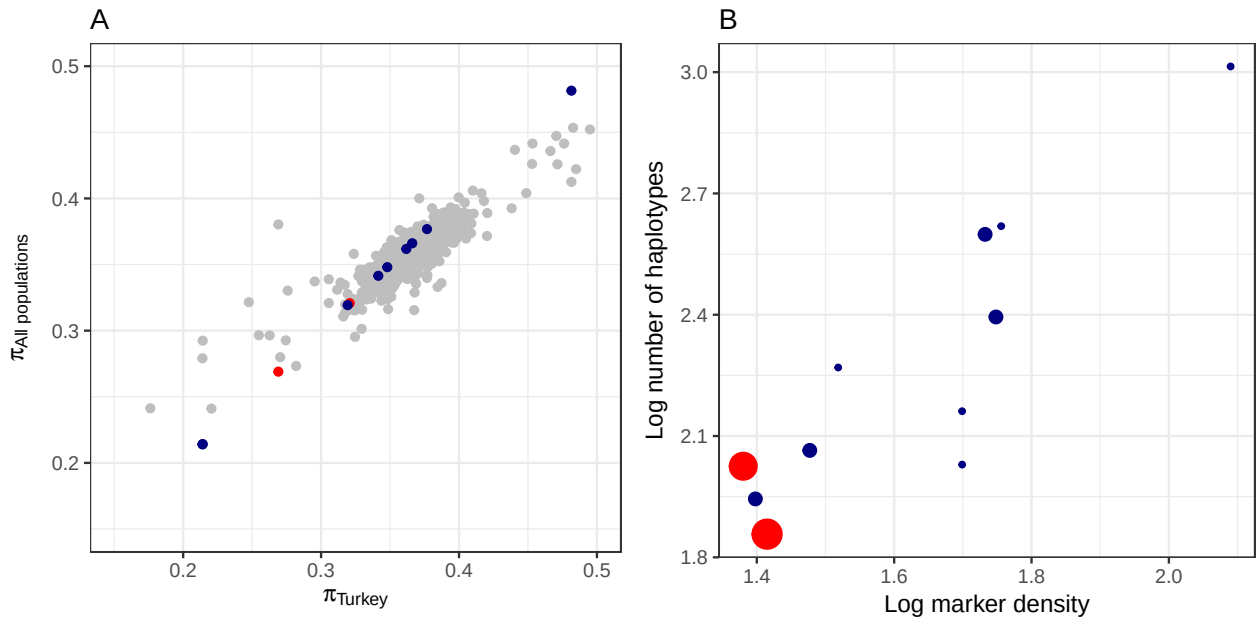

Figure S8 Genetic diversity in F_{ST} outlier regions in European great tit populations. Each point is an outlier window, with point size scaled by the number of populations the outlier was found in - "shared" haplotypes are coloured red and "unique" haplotypes dark blue (see main text for details). In **A**, Nucleotide diversity for each window is calculated for the proposed refugial population in Turkey, and plotted against the mean for all other populations. In **B**, SNP marker density is plotted against haplotype richness for each region.

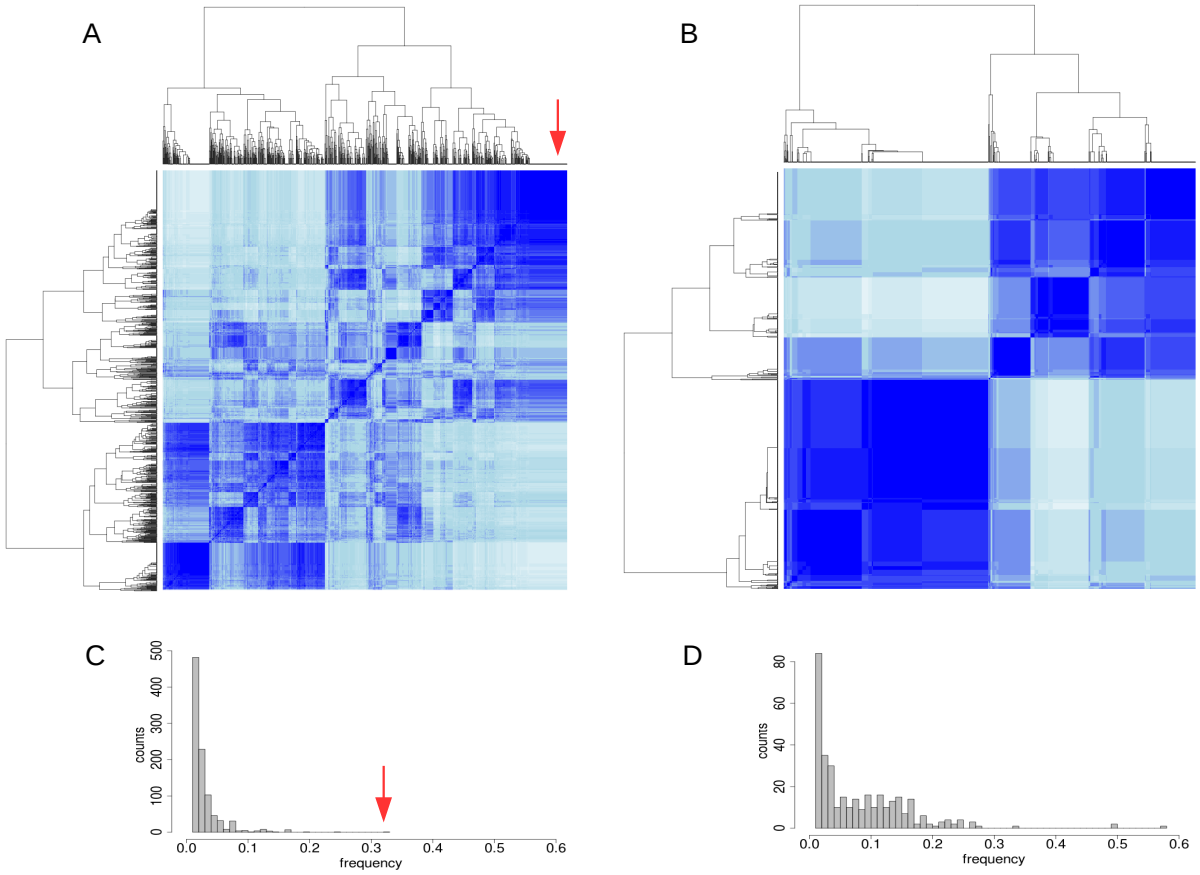

Figure S9 Haplotype structure underlying regions of high differentiation. **A** IBS clustering of all European haplotypes in the region containing a signal of high differentiation with Turkey specific to the Oulu population. The red arrow indicates the haplotype at highest frequency in Finland Colour indicates IBS similarity (range 0:white - 1: dark blue). **B** IBS clustering of all European haplotypes in the region on chromosome 2 containing a signal of high differentiation with Turkey in multiple populations. **C** Frequency of all identified haplotypes within each population. The red arrow indicates the high-frequency haplotype within Finland **D** Frequency of all identified haplotypes spanning the region on chromosome 2 within each population.